Newton's Laws with a Pulley Cart

U3 / S3.1

NAME _____ DATE _____ PERIOD _____

GROUP MEMBERS _____

GOALS

- Show that once nothing is pulling the cart, it keeps rolling at almost the same velocity (first law).
- Measure how the cart's acceleration changes with the pull and with the cart's mass, and find the cart's mass from a graph (second law).
- Measure both forces in one pull, at the same moment, and compare them (third law).

PURPOSE

You use one setup the whole lab: a cart on a level track, pulled by a string that runs over a pulley to a hanging mass. Day 1 is Parts A and B. Day 2 is Part C and the post-lab.

BACKGROUND

A force is a push or a pull. The **net force**, $\sum F$, is all the forces on an object added together, counting their directions. Newton's laws are about the net force. In this lab the falling mass pulls the string, and the string pulls the cart.

1 Newton's first law. An object keeps doing what it is doing (staying still, or moving at a steady speed in a straight line) unless a net force acts on it.

> NEWTON'S FIRST LAW
> if $\sum F = 0$, then $a = 0$ · the velocity stays the same

Part A tests it. While the mass falls, it pulls the cart and the cart speeds up. When the mass hits the floor, the string goes slack. Now nothing pulls the cart forward, so the net force on it is about zero. The first law says the cart should keep its velocity, and it does: the velocity–time graph rises, then goes flat. It is not perfectly flat, because a little friction in the wheels still pushes backward and slows the cart slightly.

2 Newton's second law. A net force makes an object accelerate. More force gives more acceleration. More mass gives less acceleration for the same force.

> NEWTON'S SECOND LAW
> $\sum F = ma$ · $N = kg \times m/s^2$

> SLOPE OF A FORCE–ACCELERATION GRAPH
> slope = $\Delta F \div \Delta a$ = mass · $N \div (m/s^2) = kg$

Part B tests it. The cart's force sensor reads the pull of the string, F, and the cart measures its own acceleration, a. A heavier hanging mass pulls harder, so the cart speeds up faster. On a graph of F (y-axis) against a (x-axis) the points make a straight line, and the slope is the cart's mass. Add a mass bar to the cart and the same pull gives less acceleration, so the line gets steeper. The line will not pass exactly through zero, because a small part of the pull only cancels friction.

3 Newton's third law. Forces come in pairs. If object A pulls on object B, then B pulls back on A with a force of the same size in the opposite direction.

> NEWTON'S THIRD LAW
> $F_{A\text{ on }B} = -F_{B\text{ on }A}$ · same size, opposite direction

Part C tests it. You hook a second force sensor, the GDX-FOR, between the cart and the string. The GDX-FOR pulls on the cart, and the cart pulls back on the GDX-FOR. Each sensor reads the pull on its own hook. By the third law, those two pulls are the same size at every moment, whether the cart is held still or moving. While the cart is held still, both should equal the weight of the hanging mass, **W = mg** (weight), with g = 9.8 m/s².

PERCENT ERROR — PART B

(| experimental – accepted | ÷ accepted) × 100

PERCENT DIFFERENCE — PART C

(| A – B | ÷ [(A + B) ÷ 2]) × 100

PRE-LAB — DUE AT THE START OF DAY 1

Answer these in your lab notebook before you walk in. Also copy the headings of Tables 1–3 (last page) into your notebook, with room for the rows you will need. Do not write on this handout.

1. State each of Newton's three laws (from the background) in your lab notebook.
2. Your pulley is 0.95 m above the floor. Use the string-length estimate in Setup step 3 to find how long your string should be, loop to loop.
3. Sketch the velocity–time graph you expect in Part A, from the moment you let go until you catch the cart. Label the point where the string goes slack.

MATERIALS — PER GROUP

- Go Direct Sensor Cart and one cart mass bar
- Dynamics Track with an end stop, and a pulley that clamps to the track end
- String, about 1.5 m, plus scissors and a meter stick
- Mass hanger and slotted masses, and a folded towel for the floor
- Balance, and a device with Graphical Analysis
- **Day 2:** Go Direct Force and Acceleration Sensor (GDX-FOR), pictured

Go Direct Force and Acceleration Sensor (photo to be added)

Go Direct Force and Acceleration Sensor

PROCEDURE

Setup

DAY 1 · ABOUT 10 MIN

1. Clamp the pulley to one end of the track, so it sits just past the edge of the bench. Put the end stop on the other end.
2. Level the track. Set the cart at rest in the middle. If it starts to roll, slide folded paper under the low end until it stays put.
3. Measure the height of the top of the pulley above the floor. Cut a piece of string and tie a small loop at each end, so the string is this long from loop to loop:

   STRING-LENGTH ESTIMATE

   string length ≈ pulley height above the floor + 40 cm

   When the mass lands, about 40 cm of string still runs from the cart to the pulley, so the cart has 40 cm to coast before it reaches the pulley.
4. Hook one loop on the cart's force-sensor hook, with the hook end of the cart facing the pulley. Run the string over the pulley and hang the mass hanger from the other loop.
5. Check the length. Roll the cart toward the pulley until the hanger touches the floor. The cart should be at least 40 cm from the pulley. If it is closer, retie one loop to shorten the string.
6. Raise or lower the pulley until the string runs level, from the hook straight to the top of the pulley.
7. Lay the folded towel on the floor under the hanger, so the mass does not bounce when it lands.

8. Mass the cart on the balance and record it in kg. Turn on the cart, connect it to Graphical Analysis, and show velocity, acceleration and force against time. Use the acceleration the cart measures with its wheel for every recorded value of a.

Before every run: lift the hanger so the string is slack, and zero the force sensor. Start each run with the cart about 1 m from the pulley, held still. Let go without pushing, and catch the cart before it reaches the pulley.

Part A — First Law: the String Goes Slack

DAY 1 · ABOUT 10 MIN

1. Take toward the pulley as positive. Put 20 g on the string (hanger included). Zero the sensor, then hold the cart at the start.
2. Start data collection and let go of the cart.
3. Watch the mass land and the cart keep rolling. Catch the cart before the pulley, then stop data collection.
4. On the velocity–time graph, find the flat part: after the mass lands and before your catch. Fit a straight line to that part only. Record the velocity at its start and its end, and the slope.
5. Repeat for 3 trials in total.

> WHAT TO LOOK FOR The graph climbs while the mass falls, then goes almost flat the moment the string goes slack. The slope of the flat part should be close to zero, and slightly negative, because friction slows the cart.

Part B — Second Law: Change the Hanging Mass

DAY 1 · ABOUT 25 MIN

1. Take toward the pulley as positive. Put 20 g on the string. Zero the sensor, hold the cart at the start, start data collection, and let go. Catch the cart.
2. Find the speeding-up stretch: the part where the mass is still falling and the force reading is steady. Leave out the moment you let go and everything after the string goes slack.
3. For that stretch, record the average force F and the average acceleration a. Both point toward the pulley, so both are positive.
4. Repeat steps 1–3 with 40, 60 and 80 g.
5. Put the mass bar on the cart. Mass the cart and bar together and record it. Repeat steps 1–3 with 40, 60 and 80 g.
6. Make a graph of F (y-axis) against a (x-axis) with both sets of points. Fit a straight line to each set and record each slope. The slope is the mass, in kg.

> WHAT TO LOOK FOR More hanging mass gives a bigger F and a bigger a. With the bar on, the same F gives a smaller a, so that line is steeper.

Part C — Third Law: Two Sensors, One Pull

DAY 2 · ABOUT 20 MIN

1. Take the mass bar off. Unhook the string from the cart.
2. Hook the GDX-FOR's hook onto the cart's force-sensor hook. Attach the string's loop to the back end of the GDX-FOR.
3. Check that the cart's hook, the GDX-FOR and the string make one straight, level line toward the pulley.
4. Connect the GDX-FOR to the same device as the cart, so both record together. Lift the hanger and zero both sensors.
5. Put 50 g on the string. Hold the cart still by the cart body, not the GDX-FOR or the string. Start data collection, keep holding for 2 s, then let go. Catch the cart before the pulley.
6. Show both force readings on the same graph. Record each sensor's average reading while the cart was held, then while it was speeding up.
7. Repeat steps 5–6 with 100 g.

> WHAT TO LOOK FOR The two force curves lie on top of each other, held or moving. The GDX-FOR pulls the cart toward the pulley. The cart pulls the GDX-FOR back, away from the pulley. Both read positive because each sensor shows a pull on its own hook as positive.

Use the rest of Day 2 for the post-lab.

DATA TABLES

Copy the headings into your notebook before Day 1. The filled rows show what a finished row looks like. Fill in your own rows **as you take each run**, with units, not from memory at the end.

TABLE 1 · PART A — FIRST LAW EXAMPLE – YOUR NUMBERS WILL DIFFER

Trial	Hanging mass (g)	Velocity at start of flat part (m/s)	Velocity at end of flat part (m/s)	Slope of flat part (m/s²)
1	*20*	*0.81*	*0.78*	*–0.094*
2	*20*	*0.83*	*0.79*	*–0.108*

Above the table, record the cart mass from the balance. *Example:* cart mass = 0.282 kg.

TABLE 2 · PART B — SECOND LAW EXAMPLE – YOUR NUMBERS WILL DIFFER

Run	Cart setup	Hanging mass (g)	Force F (N)	Acceleration a (m/s²)
1	*cart only*	*20*	*0.18*	*0.53*
2	*cart only*	*40*	*0.33*	*1.06*

Below the table, for cart only and for cart + bar: mass from the balance (kg), slope of the F–a line (kg), and percent error.

TABLE 3 · PART C — THIRD LAW EXAMPLE – YOUR NUMBERS WILL DIFFER

Run	Hanging mass (g)	Cart held or moving	Cart sensor reading (N)	GDX-FOR reading (N)	Percent difference (%)
1	*50*	*held*	*0.49*	*0.48*	*2.1*
2	*50*	*moving*	*0.27*	*0.26*	*3.8*

If you write a number wrong, draw one line through it and write the correct value beside it. Never scribble it out and never erase.

POST-LAB ANALYSIS

Answer these in your lab notebook. Show every calculation with units.

1. Explain how Part A showed Newton's first law. Use your velocity–time graph: what was the net force on the cart during the flat part, and what did its velocity do? Why was the flat part not perfectly flat? PART A
2. Explain how Part B showed Newton's second law. Use your F–a graph: what happened to a as F went up, and what happened to the line when you added the mass bar? PART B
3. Calculate the percent error of each slope. The accepted value is the balance mass: the cart alone for the first line, the cart and bar together for the second. PART B
4. Name one thing in your setup that made your Part B mass differ from the balance mass, and say which way it pushed your result. PART B
5. Explain how Part C showed Newton's third law. Use your two force readings: how did their sizes compare, and which way did each force point? PART C
6. Calculate the percent difference between the cart sensor and the GDX-FOR for each run in Table 3. PART C